Phase 1 – Foundations (Learning Basics + Mini Practice)

□ Duration: 1–2 weeks

□ Learning Goals:

- Understand what data preprocessing is and why ML models depend on it.
- Get comfortable with Python libraries:
  - pandas □ cleaning & handling missing data.
  - numpy □ numerical operations.
  - matplotlib & seaborn □ visualize data quality.

□ What to Learn (Daily small chunks):

1. Data types (categorical, numerical, time-series, text).
2. Handling missing values (mean, median, mode, interpolation).
3. Removing duplicates, handling outliers.
4. Normalization vs. Standardization.
5. Encoding categorical data (One-hot, Label encoding).
6. Feature scaling (MinMaxScaler, StandardScaler).

□ Mini-Practice (before project):

- Take a sample dataset (Titanic dataset, or Kaggle’s disease dataset).
- Try:
  - Fill missing values.
  - Normalize age/salary columns.
  - Encode categorical features like “gender”.

 Output A cleaned dataset ready for analysis.

Phase 2 – Dataset Exploration (Start Project Work)

 Duration: 1 week

 Your Tasks:

1. Collect historical disease outbreak datasets (cholera, malaria, dengue, etc.).

2. Inspect dataset:

   Which columns are useful? (rainfall, temperature, population density, cases reported).

   Which are noisy/useless? (irrelevant IDs, unneeded text).

3. Summarize dataset stats:

   .info(), .describe() in pandas.

   Missing values % per column.

 Output A Data Audit Report (what’s missing, what’s useful, what needs fixing).

Phase 3 – Cleaning & Preprocessing (Core Role)

 Duration: 2–3 weeks

□ Your Detailed Workflow:

1. Handle Missing Data
    - Replace with mean/median (for numbers).
    - Mode/most frequent (for categorical).
    - Or drop rows if too incomplete.
2. Outlier Treatment
    - Use boxplots/z-scores to detect.

- Decide: cap extreme values or remove them.

3. Feature Encoding

   - Convert “Region” □ numbers.
   - Convert “Season” □ one-hot encoding.

4. Feature Scaling

   - Apply MinMaxScaler for rainfall/temperature.
   - StandardScaler for features like “population density”.

5. Final Dataset Creation

- Save as cleaned_dataset.csv for model team.

 Output Clean dataset + Jupyter notebook documenting every step.

Phase 4 – Collaboration with Modeling Team

 Duration: Alongside Phase 3–4

 Your Tasks:

1.Share the cleaned dataset with ML teammates.

1. Get their feedback (maybe they need extra derived features).

Example:

- Convert “rainfall + season” → a new feature called “flood risk score”.

2. Iterate preprocessing if models underperform.

📌 Output → Final refined dataset + preprocessing pipeline (code).

Phase 5 – Building a Reusable Preprocessing Pipeline

⏳ Duration: 1 week

🔑 Your Extra Value Add:

Instead of cleaning manually every time, build a preprocessing pipeline using:

- scikit-learn’s Pipeline()

  Functions for missing value handling, encoding, scaling.

 Output

- Reusable pipeline code.
- Documented steps: “How raw data becomes ML-ready input.”

Phase 6 – Hackathon Delivery (Your Final Role)

 Duration: Hackathon week

 Your Role in Final Presentation:

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- Present “Data Journey”: Raw □ Clean □ ML-ready.
- Show visual before/after cleaning.
- Highlight key engineered features (like combining rainfall + season □ flood index).
- Provide dataset + pipeline to model team live.

□ Output □ Smooth handoff of high-quality, ML-ready dataset.

□ Skills You’ll Gain Along the Way

□ Practical Pandas/Numpy mastery.

□ Hands-on missing data handling & feature engineering.

□ Visualization for data quality.

□ Building preprocessing pipelines.

□ Team collaboration & documentation skills.
